Cloud Computing Lab Manual

AWS Cloud Labs – Comprehensive Guide

Institution Name

Department of Computer Science and Engineering
Course Title: Cloud Computing Lab
Course Code: CSE2273
Academic Year: 2025 – 26
Prepared By: Mr. Bailappa Bhovi
Reviewed By: Dr. Chandrashekar. V

Introduction

This Cloud Computing Lab Manual is designed to provide students with hands-on experience in various AWS Cloud services and concepts, including virtualization, cloud storage, networking, cloud security, serverless computing, and application deployment. The manual is aligned with the Outcome-Based Education (OBE) framework to ensure the development of both practical skills and conceptual understanding in cloud technologies.

Course Outcomes (COs)

After successful completion of this course, students will be able to:

- CO1: Demonstrate the ability to set up and configure various cloud computing environments and services.
- CO2: Apply virtualization techniques for creating and managing computing resources.
- CO3: Implement secure and scalable cloud applications using AWS services.
- CO4: Configure networking, storage, and database services in the cloud.
- CO5: Analyze and optimize cloud-based deployments for performance and cost-effectiveness.

List of Experiments

1. Introduction to Cloud Environments
2. Exploring Cloud Service Models (IaaS, PaaS, SaaS)
3. Comparison of Deployment Models
4. Virtual Machine Setup (Local Virtualization)
5. Creating and Managing Virtual Instances on Cloud
6. Create Virtual Network on Cloud

7. Cloud Storage Services
8. Hosting a Static Website on S3
9. Cloud Databases
10. Identity and Access Management (IAM)
11. Data Encryption and Key Management
12. Securing Applications using Amazon Cognito
13. Load Balancing and Auto Scaling
14. Deploy a Web Application using PaaS
15. Serverless Computing with AWS Lambda

Lab 1: Introduction to Cloud Environments

Objective(s)

- Understand AWS as a cloud service provider.
- Learn to create and configure an AWS Free Tier account.
- Explore AWS Management Console.
- Create and configure an IAM user for secure access.

Theory / Background

AWS provides a suite of scalable cloud services. Before working on AWS, users must register for an account, understand the console layout, and create IAM (Identity and Access Management) users for security best practices. Using IAM ensures the root account credentials are not used for daily operations, reducing security risks.

Software/Tools Required

- AWS Free Tier account (https://aws.amazon.com/free/)
- Web Browser (Google Chrome / Firefox)
- AWS CLI (optional, for command-line operations)

Procedure / Steps

Step 1: Create AWS Free Tier Account

1. Go to https://aws.amazon.com/free/
2. Click **Create an AWS Account**.

3. Enter email, password, and account name.
4. Provide contact and payment information (a valid credit/debit card is required; charges apply only for usage beyond Free Tier limits).
5. Verify phone number and identity.
6. Select the **Basic Plan (Free)**.
   📸 *Screenshot Point:* After successful account creation.

Step 2: Explore AWS Management Console

1. Log in at https://aws.amazon.com/console/.
2. Familiarize yourself with:
   - **Service Search Bar**
   - **Pinned Services**
   - **Region Selector** (top right)
   - **Account Menu**
   - **Support Menu**
     📸 *Screenshot Point:* Highlight the Services menu.

Step 3: Create an IAM User

1. Navigate to **IAM** service from the console.
2. Click **Users → Add User**.
3. Enter **username** (e.g., lab-user1).
4. Select **Access type**:
   - **AWS Management Console access**
   - **Programmatic access** (for CLI/SDK use)
5. Create a custom password or auto-generate.
6. Assign permissions:
   - Attach policies directly (e.g., AdministratorAccess for labs).
7. Review and create user.
8. Save the **Access Key ID** and **Secret Access Key** (for CLI use).
   📸 *Screenshot Point:* IAM user summary page.

Sample Code / Commands *(AWS CLI)*

```
# Configure AWS CLI with new IAM user
aws configure
# Enter Access Key, Secret Key, Region, and Output format

# List all S3 buckets (will be empty initially)
aws s3 ls
```

Observations

Step Performed	Output / Observation
AWS Account Created	Account ID: _____
IAM User Created	User ARN: _____
AWS CLI Configured	Verified Access

Result

Successfully created an AWS Free Tier account, explored the AWS Management Console, and configured an IAM user with access credentials.

Viva Questions

1. What are the benefits of using IAM users instead of the root account?
2. What is the AWS Free Tier and what does it offer?
3. Name at least three core AWS service categories.
4. How does AWS region selection affect your resources?
5. Explain the difference between AWS CLI and AWS Management Console.

Assessment Criteria

- **Account Creation**: AWS account successfully created.
- **Console Navigation**: Ability to identify and access AWS services.
- **IAM Configuration**: Correctly configured IAM user with permissions.
- **Security Best Practices**: Avoided use of root account for daily operations.
- **Documentation**: Submitted screenshots and filled observation table.